Inductor
switches
High side
Low side
Capacitor
Driver (if required:why logic mosfet cant be used)~
input capacitor
The control loop(LC-for 1kHz step-expect to tune the compensator.)
Bootstrap Circuit
Current sensing

Specs chosen:
10W converter,CCM
Vin=12-24V
Vout=5V
fsw=37kHz with +/-3kHz and 1kHz step

Mosfet selection
VDS
ID
RDS

Vbreakdown=12*margin(0.2 or 0.3)=16V standard =20V

Vin=12V ,10W Iin (it is avg!)=10/12=0.834A
Vout=5V,10W,Iout=2A
IDS(low side)=6A to 12 A(select between safety factor of 3, 6)

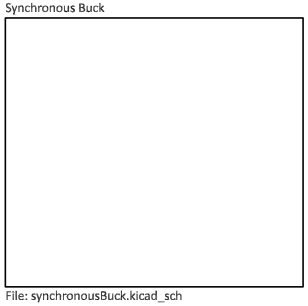
losses=switching+conduction

Pcond=Ids^2*Rds
Assuming an efficiency of 95% of 10W out,input=10.52W
losses allowed=526.3mW
assuming 1/3 losses from 2 mosfets
Ploss=87.8mW=>Pcond=Ploss/2
Pcond=Ids^2*Rds
acceptable Rds=63.056mOhm

Higher Ids->lower Rds->bigger die->larger gate capacitance
Gate drive power per FET is Qg × Vgs × fsw:
15 nC × 5 V × 37 kHz ~ 2.71 mW

Higher Vds->higher Rds

Synchronous Buck



File: synchronousBuck.kicad_sch

Iout=10W/5V=2A
deltaIL=Iout*0.40
D=5/12*n

$$L_1 = \frac{V_{out} \cdot (1-D)}{f_{sw} \cdot \Delta I_{L1}}$$

L1=((5)*(1-0.4166))/(37-3kHz)(0.40*2A)
=>L1=(5*0.5834)/34kHz * (0.8)
L1=107uH
L1≥107uH

L1=((5)*(1-0.5))/(37kHz)(0.40*2A)
=>L1=84.5uH

Inductor saturates at Imax and IL <Imax
Inductor peak current:
Ipk=Iout+delta(I)/2

output Capacitor seletion:
1.filters inductor current at output
2.supports load support before regulator can react

Filter criteria 1 (based on filter inductor current ripple for n phases):

$\Delta I_{Cout} = \frac{V_{out}}{n \cdot f_{sw}} \cdot \frac{1-D}{L}$

delta(ICout)=(12V/1*37kHz)*((1-0.4166)/120uH)
deltaIcout=1.577A

$$\Delta Q_{Cout} = \frac{T_{sw}}{n \cdot 2} \cdot \frac{1}{2} \cdot \frac{\Delta I_{Cout}}{2} = \frac{\Delta I_{Cout}}{8 \cdot n \cdot f_{SW}}$$

delta(QCout)=1.577/(8*1*37kHz)
delta(QCout)=5.3278uC

$$C_{out} = \frac{1-D}{\frac{\Delta V_{Cout}}{V_{out}} \cdot 8\,L \cdot (n \cdot f_{SW})^2}$$

minimum output capacitance for 1st crieteria(Cout):
Cout=(1-0.411)/(delta(VCout)/5V)*8*120uH*(1*34kHz)^2
Cout=0.5884/((delta(VCout)/5V)*0.00096*(1156M)

(delta(VCout) taken as 1% of Vout: 5*0.01=50mV)(considered fmin=34kHz)
Cout=0.5884/((50mV/5V)*0.00096*(1156M)
Cout=53.02uF

ESR ripple=ΔIL × ESR
should be less than
50mV>0.8*ESR

Critreria 2 not considered
(assuming constant load)

if i choose 200uH
L1=((5)*(1-0.4166))/(37-3kHz)(0.40*2A)
=>200uH=(5*0.5834)/34kHz * (delta(IL1)
delta(IL1)=0.4289A
21.3% of I out

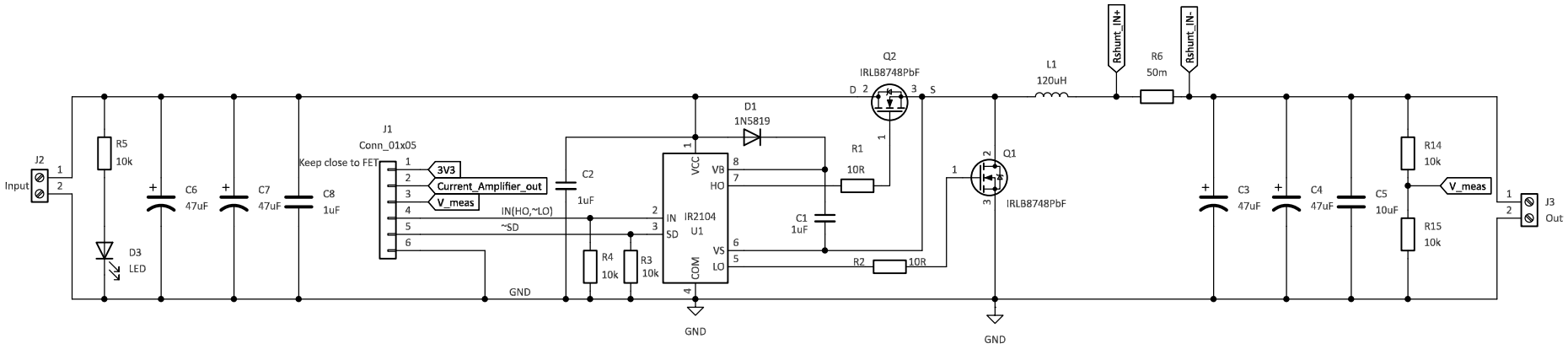
Note: for CCM
A synchronous half-bridge with a gate driver, not a diode. This is what makes forced CCM possible at all. With a Schottky, the board goes DCM below about 0.3-0.36 A and you can't fix it in firmware.
Bipolar current sensing. The sense amp needs a mid-supply bias (about 1.65 V on a 3.3 V ADC) so it can read negative inductor current at light load.
Inductor footprint with some flexibility. Choose a footprint that fits both 100 µH and 120 µH parts, or a bit larger, in case you want to swap values later.

minimum output capacitance for 1st crieteria(Cout)(for 200uH inductor):
Cout=53.02uF(120/200)
Cout=31.812uF

Current_Amplifier

File: Current_Amp.kicad_sch

(IL)max=2A
Vdrop=2A*R
Input common mode voltage
Vcm=avg(Vi+Vinv)
high side or low side
High side(in this case)(Vcm~Vs)
low side:sensing load->shunt->gnd (Vcm~0)



1/4Watt 1206 Resistors rated CHECK THIS

shows the basic connections for the INA199. The input pins, IN+ and IN-, must be connected as close as possible to the shunt resistor to minimize any resistance in series with the shunt resistor.

Add for convenience:
Test points on the switch node, both gates, VB, VS, Vout and GND.
Mounting holes.
A fuse or reverse-polarity protection on J2, if you want it. A wrong connection at 12 V would destroy the board.

Vshunt =Rshunt*Iout
Vshunt=2A*0.05
Vshunt=100mV

Vref=mid of (Vsupply)=3.3V/2=1.65V

$$Gain_{Diff_Amp} = \frac{V_{OUT_MAX} - V_{OUT_MIN}}{R_{SHUNT} \times (I_{MAX} - I_{MIN})} = \frac{3.2V - 100mV}{100m\Omega \times (1A - (-1A))} = 15.5\frac{V}{V}$$

Gain=(3.3V-100mV)/(50mOhm*(2A-0))
Gain=32V/V
Rf/Rin=32/1

